

Exp 7

Procedure to Transfer Files from One Virtual Machine to Another Virtual Machine

Aim:

To transfer a file from one Ubuntu virtual environment (VM1) to another Ubuntu virtual environment (VM2) using the SCP (Secure Copy) command.

Prerequisites:

- Windows system with WSL 2 and Ubuntu installed.
- Two Ubuntu instances: Ubuntu (VM1) and Ubuntu2 (VM2).
- OpenSSH Server installed and running on VM2.

Procedure:

Step 1: Start Ubuntu1

Open Ubuntu1 from Windows CMD using: `wsl -d Ubuntu`

Step 2: Create a test file

`echo "File transferred from VM1" > vm1.txt`

Step 3: Verify the file

`cat vm1.txt`

Step 4: Start Ubuntu2

Open a separate Windows CMD window and run: `wsl -d Ubuntu2`

Step 5: Start SSH Server

In Ubuntu2, run: `sudo service ssh start`

Step 6: Check SSH status

Run: `sudo service ssh status` and confirm Active: active (running).

Step 7: Transfer the file

In Ubuntu1, run: `scp vm1.txt administrator@192.168.96.127:/home/administrator/`

Step 8: Verify the file

In Ubuntu2, run: `cat /home/administrator/vm1.txt`

Commands Used:

```
wsl -d Ubuntu
echo "File transferred from VM1" > vm1.txt
cat vm1.txt
wsl -d Ubuntu2
sudo service ssh start
sudo service ssh status
scp vm1.txt administrator@192.168.96.127:/home/administrator/
cat /home/administrator/vm1.txt
```

Output:

The file vm1.txt was successfully transferred from Ubuntu VM1 to Ubuntu2 using SCP. The received file was verified in Ubuntu2 and displayed: "File transferred from VM1".

```
256 SHA256:1/b9w8NZqq2qY7FNxDGKWT8nx/FDN6Jsd5BzNCpKV8Q root@SriBalaji (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:XzVy0vpgT428q+ioz8S4T+ET6q8ASCaq0daqlJaP0o root@SriBalaji (ED25519)
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
administrator@SriBalaji:/mnt/c/Users/Administrator$ sudo service ssh start
administrator@SriBalaji:/mnt/c/Users/Administrator$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-09-15 07:49:57 UTC; 9s ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 1379 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 1381 (sshd)
    Tasks: 1 (limit: 9405)
   Memory: 1.2M (peak: 1.6M)
      CPU: 18ms
   CGroup: /system.slice/ssh.service
           └─1381 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 15 07:49:57 SriBalaji systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 15 07:49:57 SriBalaji sshd[1381]: Server listening on 0.0.0.0 port 22.
Sep 15 07:49:57 SriBalaji sshd[1381]: Server listening on :: port 22.
Sep 15 07:49:57 SriBalaji systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
administrator@SriBalaji:/mnt/c/Users/Administrator$ |
```

Figure 1: SSH server running successfully in Ubuntu2

```
administrator@SriBalaji:/mnt/c/Users/Administrator$ echo "File transferred from VM1" > vm1.txt
administrator@SriBalaji:/mnt/c/Users/Administrator$ cat vm1.txt
File transferred from VM1
administrator@SriBalaji:/mnt/c/Users/Administrator$ scp vm1.txt administrator@192.168.96.127:/home/administrator/
ssh: connect to host 192.168.96.127 port 22: Connection refused
scp: Connection closed
administrator@SriBalaji:/mnt/c/Users/Administrator$ |
```

Figure 2: SCP file transfer completed successfully

```
bin$
ste 1 administrator@SriBalaji:/mnt/c/Users/Administrator$ cat /home/administrator/vm1.txt
cat: /home/administrator/vm1.txt: No such file or directory
E Pa administrator@SriBalaji:/mnt/c/Users/Administrator$ exit
re Li logout
ect 3 C:\Users\Administrator>wsl -d Ubuntu2
'Gra wsl: Failed to translate 'D:\New Downloads\bin'
view wsl: Failed to translate 'D:\nodejs\'
wsl: Failed to translate 'D:\Antigravity\bin'
a Co administrator@SriBalaji:/mnt/c/Users/Administrator$ cat /home/administrator/vm1.txt
bin: File transferred from VM1
Join administrator@SriBalaji:/mnt/c/Users/Administrator$ cat /home/administrator/vm1.txt
File transferred from VM1
Join administrator@SriBalaji:/mnt/c/Users/Administrator$ |
```

Figure 3: Transferred file verified in Ubuntu2

Result:

Thus, the file vm1.txt was successfully transferred from one Ubuntu virtual environment (VM1) to another Ubuntu virtual environment (VM2) using the SCP command and the transferred file was verified successfully.